Automated Database Backup & Download Plan

800 CarGuru — Daily Backup to Local PC

Date: April 2, 2026 **Server:** SMF_NEW_CG (Ubuntu 24.04, MariaDB) **Server IP:** 13.220.57.16 **Database:** carguru2 **Local Destination:** D:\DB_BKP\

Overview

```
┌─────────────────┐     2 AM Daily       ┌─────────────────┐
│                 │ ──────────────────►  │                 │
│   VPS Server    │   mysqldump + gzip   │  /home/ubuntu/  │
│   (MariaDB)     │                      │  DB_BKP/        │
│                 │                      │                 │
└─────────────────┘                      └────────┬────────┘
                                                  │
                                        3 AM Daily (SCP)
                                                  │
                                         ┌────────▼────────┐
                                         │                 │
                                         │  D:\DB_BKP\     │
                                         │  (Local PC)     │
                                         │                 │
                                         └─────────────────┘
```

Daily Schedule

Time	Action	Where
2:00 AM	Database dump + compress	VPS Server
3:00 AM	Download latest backup via SCP	Local PC

Part 1: Server-Side Setup

Step 1.1: Create Backup Directory

```
mkdir -p /home/ubuntu/DB_BKP
```

Step 1.2: Create Backup Script

```
sudo nano /home/ubuntu/db_backup.sh
```

Paste:

```bash
#!/bin/bash

# ============================================
# DAILY DATABASE BACKUP SCRIPT
# Runs at 2 AM via cron
# ============================================

BACKUP_DIR="/home/ubuntu/DB_BKP"
DATE=$(date +%Y-%m-%d_%H-%M-%S)
LOG_FILE="/var/log/db_backup.log"

mkdir -p $BACKUP_DIR

echo "$(date) - Starting backup..." >> $LOG_FILE

# Dump and compress
mysqldump -u root carguru2 | gzip > "$BACKUP_DIR/carguru2_$DATE.sql.gz"

if [ $? -eq 0 ]; then
    SIZE=$(du -h "$BACKUP_DIR/carguru2_$DATE.sql.gz" | cut -f1)
    echo "$(date) - Backup completed: carguru2_$DATE.sql.gz ($SIZE)" >> $LOG_FILE
else
    echo "$(date) - ERROR: Backup failed!" >> $LOG_FILE
fi

# Keep only last 7 days of backups
find $BACKUP_DIR -name "*.sql.gz" -mtime +7 -delete
echo "$(date) - Old backups cleaned up" >> $LOG_FILE
```

Step 1.3: Set Permissions

```
sudo chmod +x /home/ubuntu/db_backup.sh
```

Step 1.4: Create Log File

```
sudo touch /var/log/db_backup.log
sudo chmod 666 /var/log/db_backup.log
```

Step 1.5: Schedule Cron Job (Daily 2 AM)

```
sudo crontab -e
```

Add:

```
0 2 * * * /home/ubuntu/db_backup.sh
```

Step 1.6: Test Manually

```
sudo /home/ubuntu/db_backup.sh
ls -lh /home/ubuntu/DB_BKP/
cat /var/log/db_backup.log
```

Server-Side Checklist

- ☑ Backup directory created
- ☑ Backup script created and executable
- ☑ Log file created
- ☑ Cron job scheduled at 2 AM
- ☑ Manual test successful

Part 2: Local PC Setup (Windows)

Step 2.1: Prerequisites

- ☑ SSH key file: `D:\DB_BKP\id_rsa_new.pem`
- ☑ Git Bash installed (provides SCP): `C:\Program Files\Git\`
- ☑ Destination folder: `D:\DB_BKP\`

Step 2.2: Fix SSH Key Permissions (One-Time)

Run in **PowerShell as Administrator**:

```
icacls "D:\DB_BKP\id_rsa_new.pem" /inheritance:r /grant:r "ABC:R"
icacls "D:\DB_BKP\id_rsa_new.pem" /remove "NT AUTHORITY\Authenticated Users"
icacls "D:\DB_BKP\id_rsa_new.pem" /remove "BUILTIN\Users"
icacls "D:\DB_BKP\id_rsa_new.pem" /remove "Everyone"
```

Step 2.3: Create Download Script

Create file: `D:\DB_BKP\download_backup.bat`

```
@echo off

REM ============================================
REM DAILY DATABASE BACKUP DOWNLOAD
```

```
REM Downloads latest backup from VPS to local PC
REM Uses Git Bash SCP
REM ============================================

set KEY=D:\DB_BKP\id_rsa_new.pem
set SERVER=ubuntu@13.220.57.16
set DEST=D:\DB_BKP\
set LOG=%DEST%download_log.txt
set SCP="C:\Program Files\Git\usr\bin\scp.exe"
set SSH="C:\Program Files\Git\usr\bin\ssh.exe"

echo ---------------------------------------- >> "%LOG%"
echo %date% %time% - Starting download... >> "%LOG%"

REM Get latest filename from server
%SSH% -i "%KEY%" %SERVER% "ls -t /home/ubuntu/DB_BKP/*.sql.gz | head -1" >
"%DEST%latest.txt"

set /p LATEST=<"%DEST%latest.txt"

REM Download only the latest backup
%SCP% -i "%KEY%" %SERVER%:%LATEST% "%DEST%"

if %ERRORLEVEL% EQU 0 (
    echo %date% %time% - Download successful: %LATEST% >> "%LOG%"
) else (
    echo %date% %time% - Download FAILED >> "%LOG%"
)

del "%DEST%latest.txt" 2>nul

echo ---------------------------------------- >> "%LOG%"
```

Step 2.4: Test Manually

- Double-click `download_backup.bat`
- Check `D:\DB_BKP\` for the `.sql.gz` file
- Check `D:\DB_BKP\download_log.txt` for success message

Local PC Checklist

- ☑ Download script created
- ☑ Key permissions fixed
- ☑ Manual test successful
- ☑ Latest backup file downloaded to D:\DB_BKP\

Part 3: Windows Task Scheduler Setup

Step 3.1: Open Task Scheduler

- Press `Win + S` → search **Task Scheduler** → Open

Step 3.2: Create New Task

1. Click **Create Basic Task**
2. **Name:** `CG2 DATABASE BACKUP`
3. **Description:** `Downloads daily database backup from VPS to D:\DB_BKP\`
4. Click **Next**

Step 3.3: Set Trigger

1. Select **Daily**
2. **Start time:** `3:00:00 AM`
3. **Recur every:** `1` day
4. Click **Next**

Step 3.4: Set Action

1. Select **Start a program**
2. **Program/script:** Browse to `D:\DB_BKP\download_backup.bat`
3. Click **Next**

Step 3.5: Finish and Configure

1. Click **Finish**
2. Right-click the new task → **Properties**
3. **General** tab:
   - Select: **Run only when user is logged on**
   - Check: **Run with highest privileges**
4. **Settings** tab:
   - Check: **Run task as soon as possible after a scheduled start is missed**
   - Check: **If the task fails, restart every** → `10 minutes`
   - Set: **Attempt to restart up to** → `3` times
5. Click **OK**

Task Scheduler Checklist

- ☑ Task created: `CG2 DATABASE BACKUP`
- ☑ Trigger set to daily 3 AM
- ☑ Action points to download_backup.bat
- ☑ Run only when user is logged on: enabled
- ☑ Run with highest privileges: enabled

Verification & Monitoring

Daily Checks (First Week)

On VPS:

```
# Check backup exists
ls -lh /home/ubuntu/DB_BKP/

# Check backup log
tail -10 /var/log/db_backup.log
```

On Local PC:

- Open `D:\DB_BKP\` — new `.sql.gz` file should appear daily
- Check `D:\DB_BKP\download_log.txt` for success entries

Weekly Checks (Ongoing)

Check	How	Expected
Backups on VPS	`ls /home/ubuntu/DB_BKP/`	7 files (last 7 days)
Backup log	`cat /var/log/db_backup.log`	No errors
Local backups	Open `D:\DB_BKP\`	Files downloading daily
Download log	Open `download_log.txt`	No failures
Disk space (VPS)	`df -h`	Not filling up
Disk space (local)	Check D: drive	Not filling up

Troubleshooting

Backup not created on VPS

```
# Check cron is running
sudo systemctl status cron

# Check cron log
grep "db_backup" /var/log/syslog | tail -5

# Run manually to see errors
sudo /home/ubuntu/db_backup.sh
```

Download fails on local PC

Issue	Solution
"Permission denied"	Check key file path and permissions
"error in libcrypto"	Re-export key from MobaKeyGen as OpenSSH format
"Connection refused"	Check server is running, check IP

Issue	Solution
"Host key verification"	Run SCP manually once to accept the key
Task doesn't run	Check Task Scheduler history for errors
"User account restriction"	Set task to "Run only when user is logged on"
PC was off at 3 AM	Enable "Run task as soon as possible after a scheduled start is missed"

Restore from backup

```
# Upload backup to server (from Git Bash)
"C:\Program Files\Git\usr\bin\scp.exe" -i "D:\DB_BKP\id_rsa_new.pem"
"D:\DB_BKP\carguru2_2026-04-02.sql.gz" ubuntu@13.220.57.16:/tmp/

# On server — restore
gunzip /tmp/carguru2_2026-04-02.sql.gz
sudo mysql -e "DROP DATABASE carguru2; CREATE DATABASE carguru2;"
sudo mysql carguru2 < /tmp/carguru2_2026-04-02.sql
```

Local Backup Cleanup (Optional)

To avoid filling up your D: drive, add this line before the last line in `download_backup.bat`:

```
REM Delete local backups older than 30 days
forfiles /P "D:\DB_BKP" /S /M *.sql.gz /D -30 /C "cmd /c del @file" 2>nul
```

Summary

Component	Location	Schedule	Retention
Database dump	VPS `/home/ubuntu/DB_BKP/`	Daily 2 AM	7 days
Local download	`D:\DB_BKP\`	Daily 3 AM	Latest only
Backup log	VPS `/var/log/db_backup.log`	Continuous	—
Download log	`D:\DB_BKP\download_log.txt`	Continuous	—

Quick Reference

Item	Value
Server IP	`13.220.57.16`
SSH Key	`D:\DB_BKP\id_rsa_new.pem`

Item	Value
SSH User	`ubuntu`
Database	`carguru2`
SCP Tool	`C:\Program Files\Git\usr\bin\scp.exe`
SSH Tool	`C:\Program Files\Git\usr\bin\ssh.exe`
VPS Backup Path	`/home/ubuntu/DB_BKP/`
Local Backup Path	`D:\DB_BKP\`
VPS Backup Script	`/home/ubuntu/db_backup.sh`
Local Download Script	`D:\DB_BKP\download_backup.bat`
Task Scheduler Name	`CG2 DATABASE BACKUP`
VPS Cron Time	2:00 AM daily
PC Download Time	3:00 AM daily
VPS Retention	7 days
Local Retention	30 days (optional)

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